

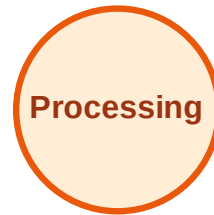
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



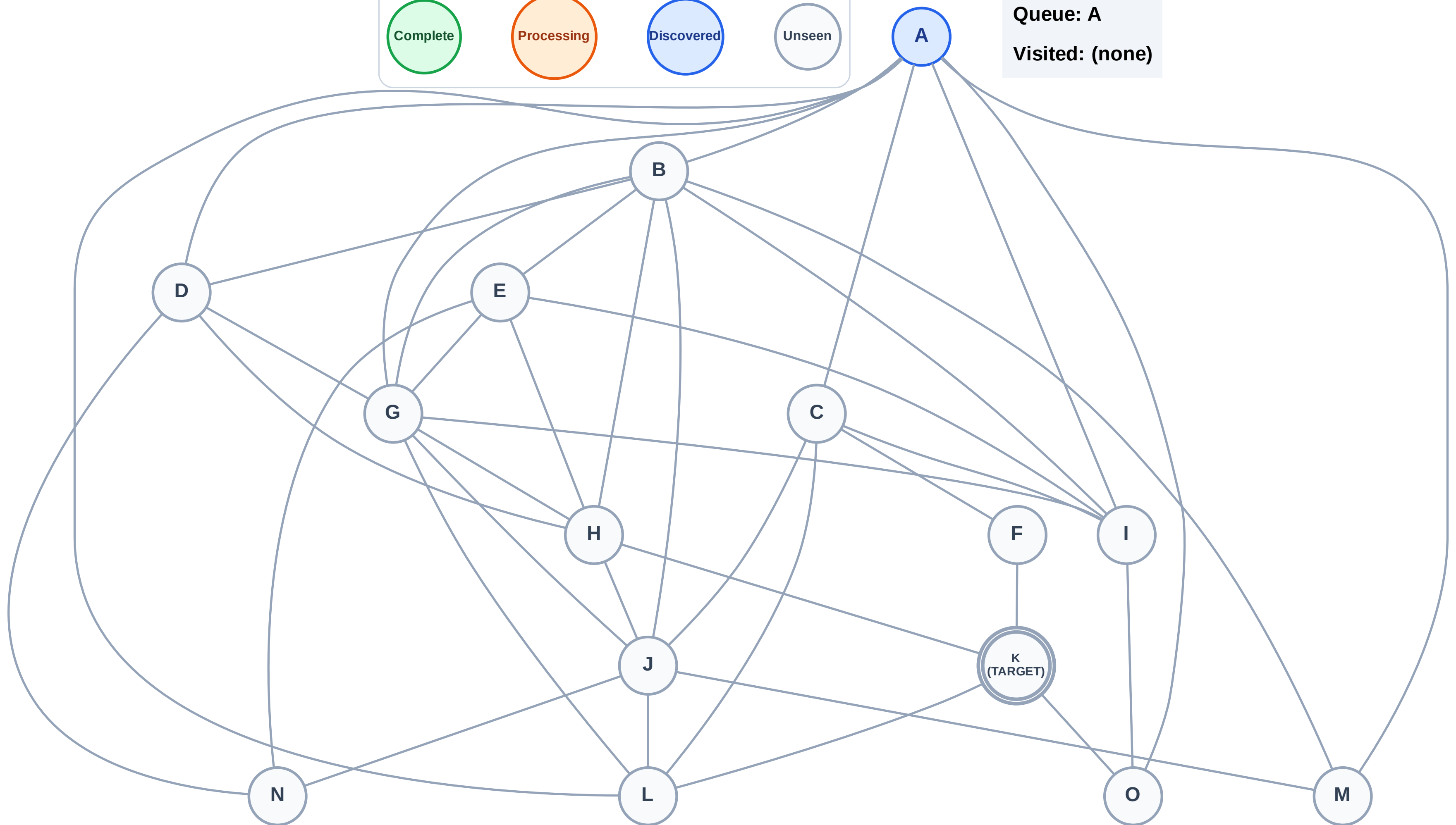
Discovered



Unseen

Queue: A

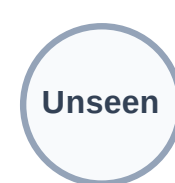
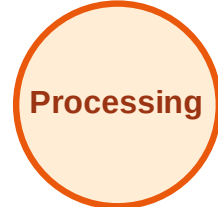
Visited: (none)



BFS Traversal

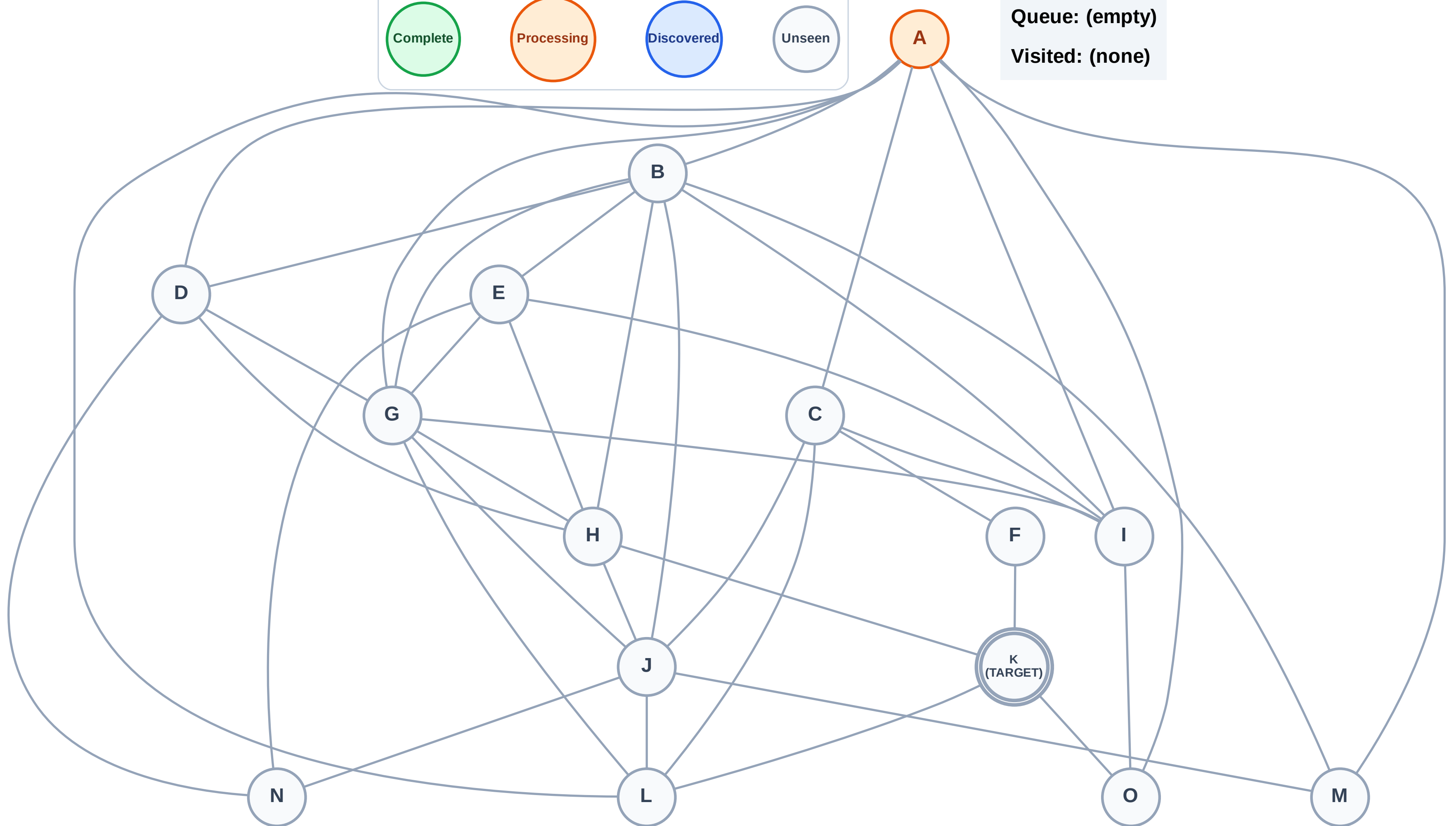
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



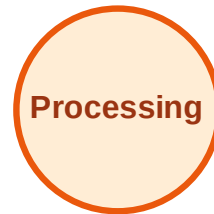
BFS Traversal

Step 3: Discover B, C, D, G, I, L, M, O from A

Legend



Complete



Processing



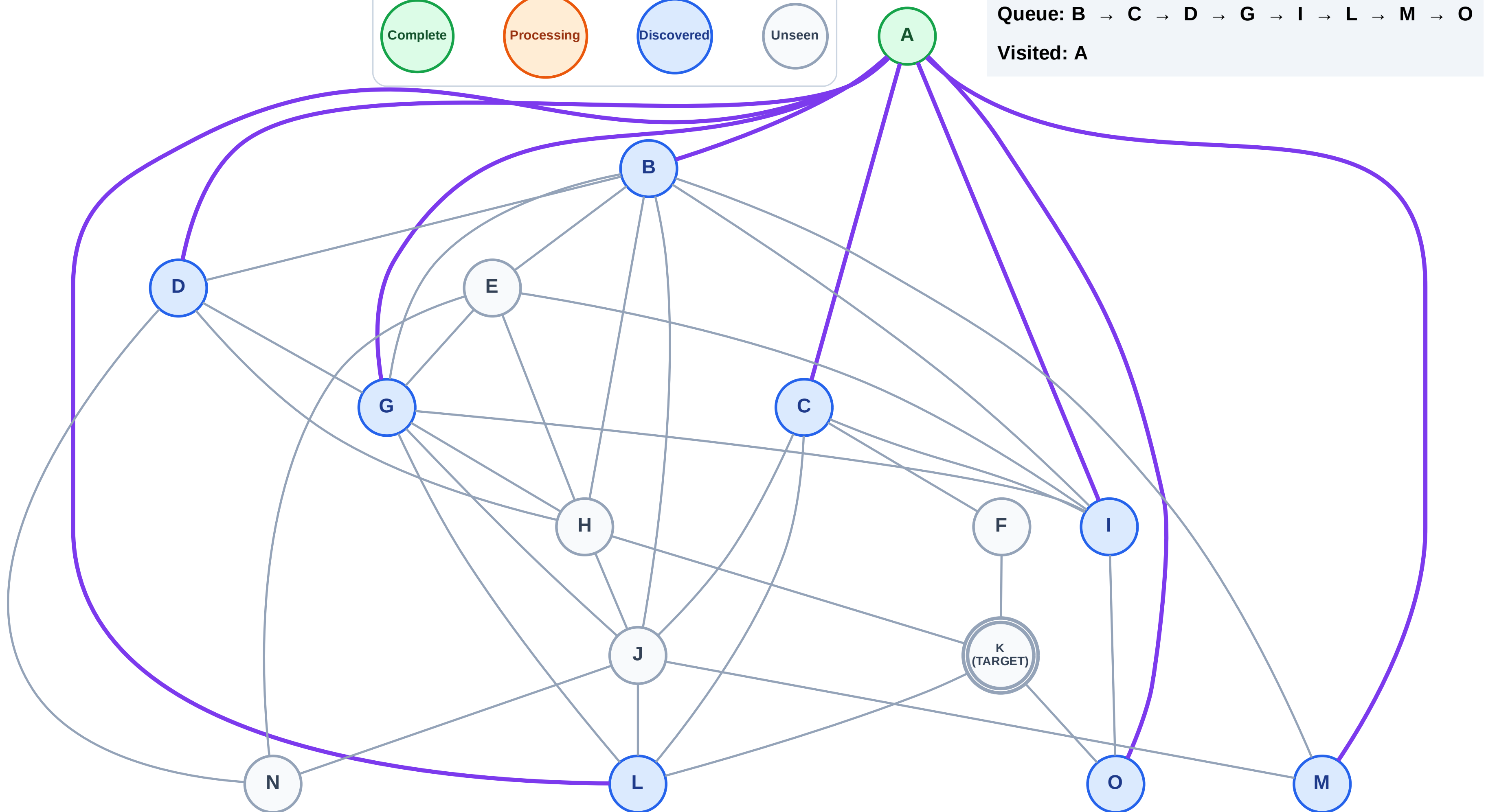
Discovered



Unseen

Queue: B → C → D → G → I → L → M → O

Visited: A



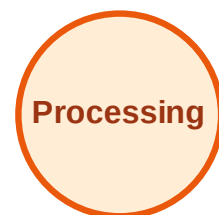
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



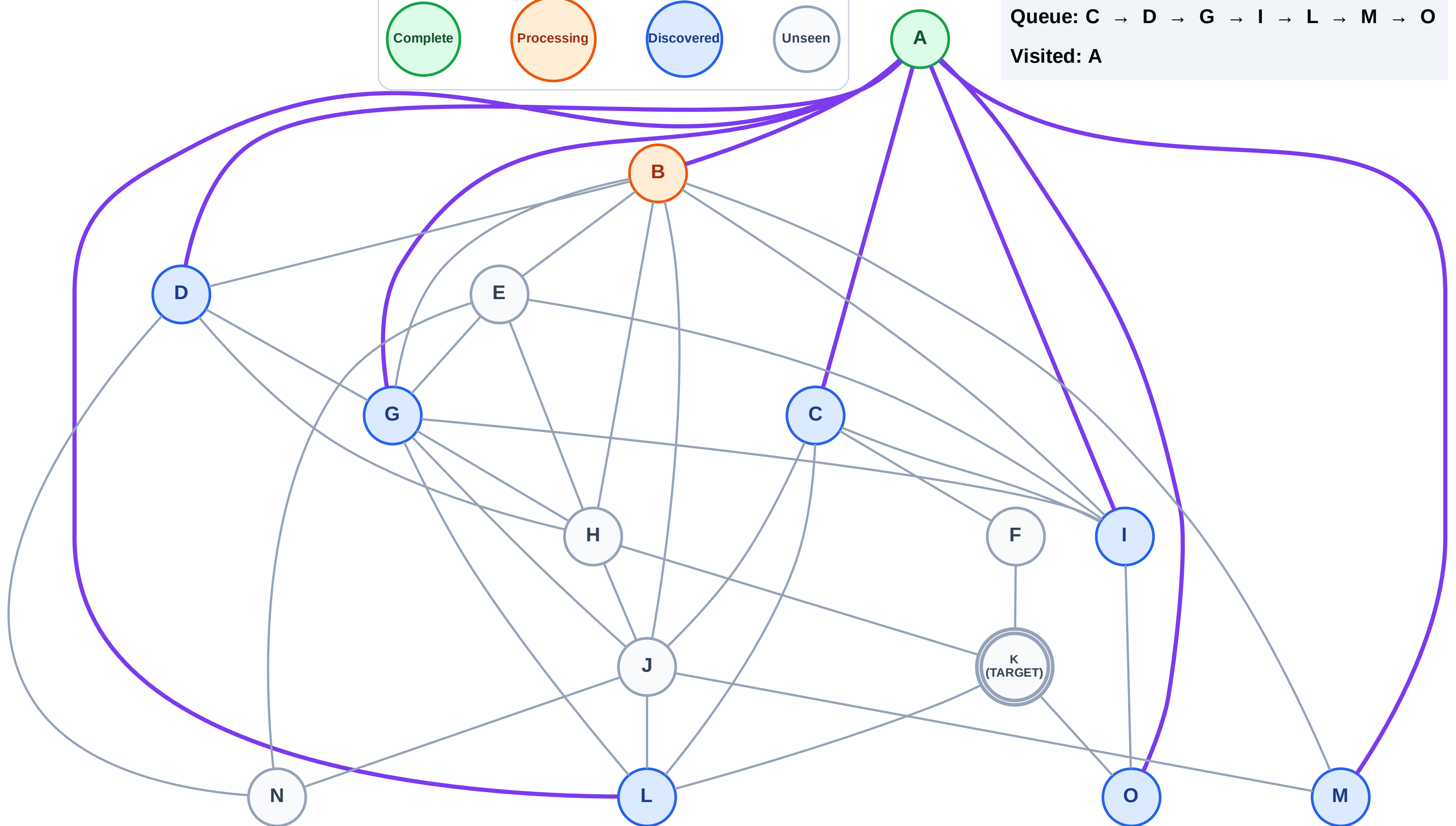
Discovered



Unseen

Queue: C → D → G → I → L → M → O

Visited: A



BFS Traversal

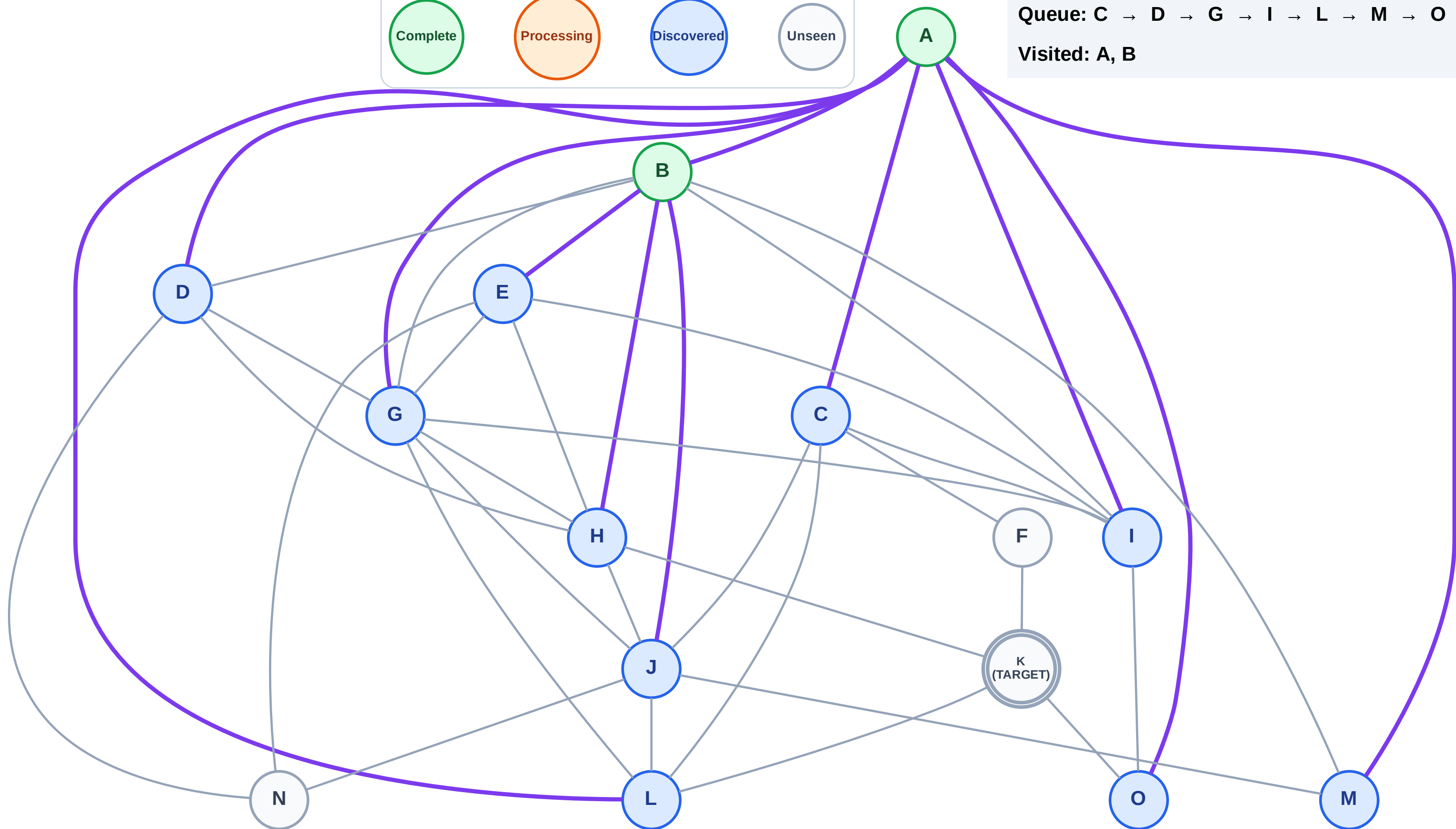
Step 5: Discover E, H, J from B

Legend



Queue: C → D → G → I → L → M → O → E → H → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

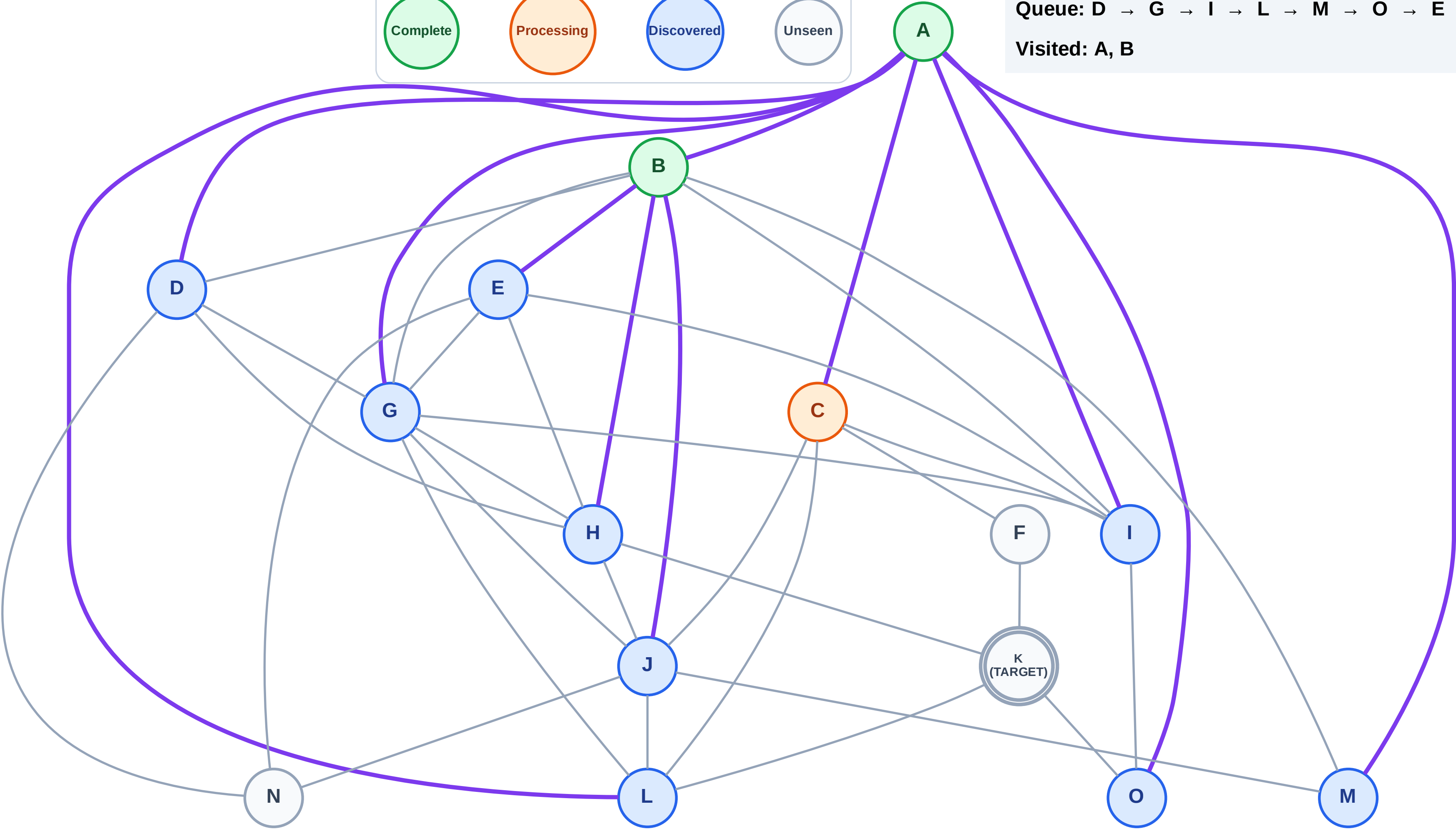
Processing

Discovered

Unseen

Queue: D → G → I → L → M → O → E → H → J

Visited: A, B



BFS Traversal

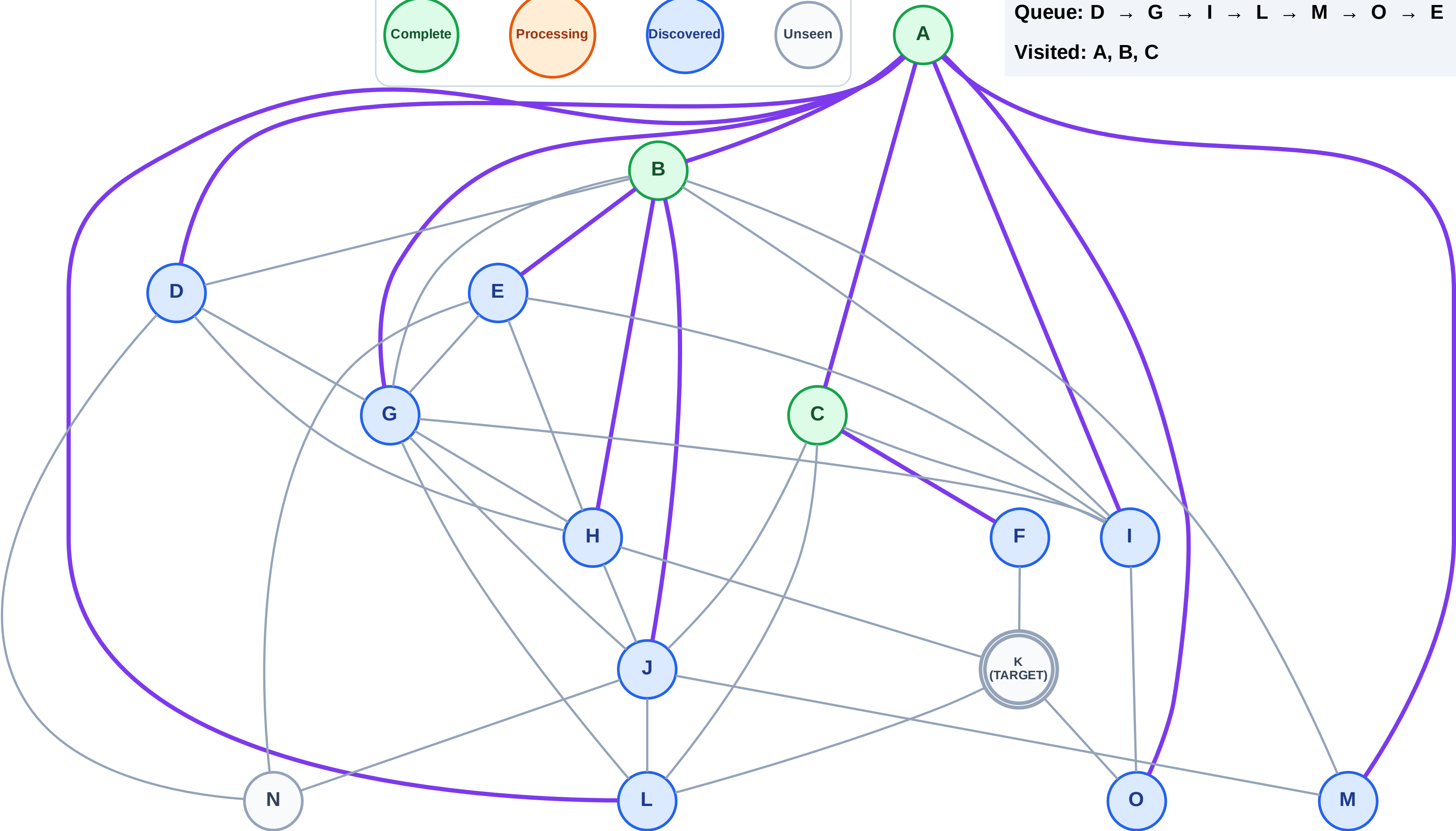
Step 7: Discover F from C

Legend



Queue: D → G → I → L → M → O → E → H → J → F

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

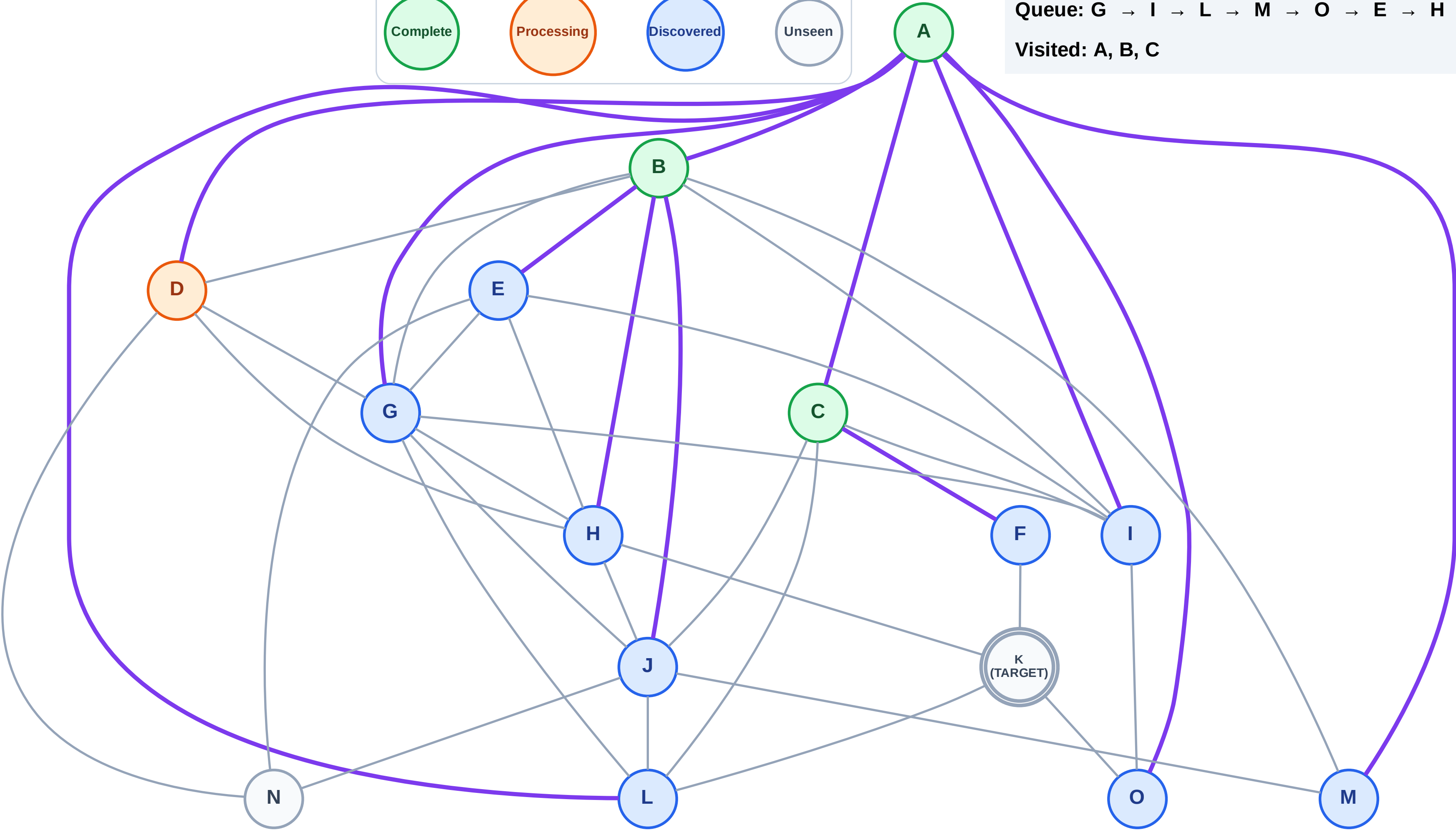
Processing

Discovered

Unseen

Queue: G → I → L → M → O → E → H → J → F

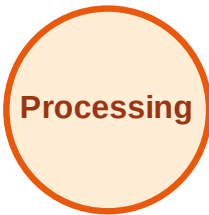
Visited: A, B, C



BFS Traversal

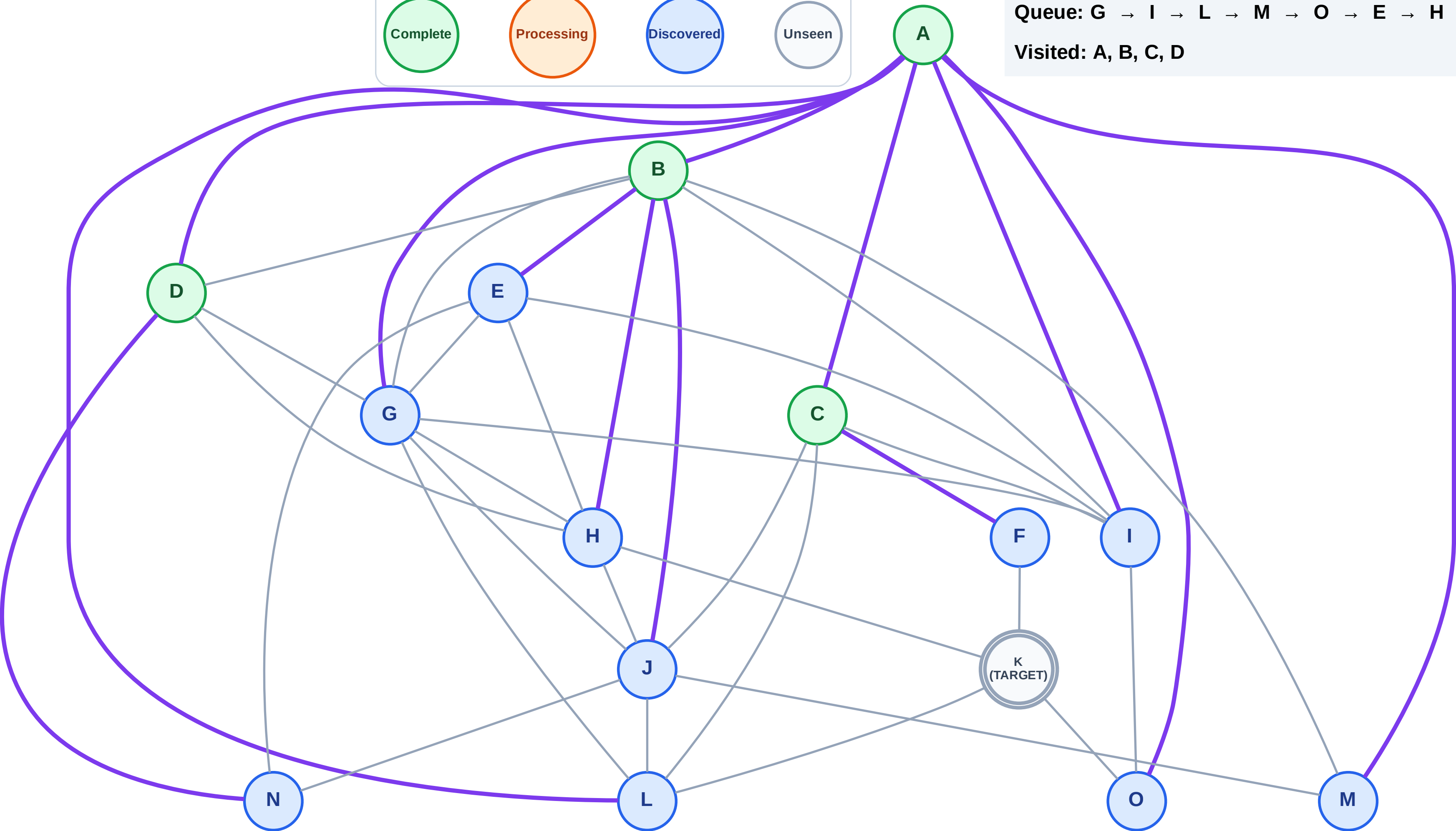
Step 9: Discover N from D

Legend



Queue: G → I → L → M → O → E → H → J → F → N

Visited: A, B, C, D



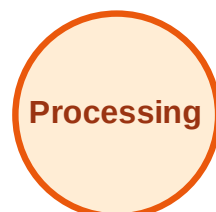
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



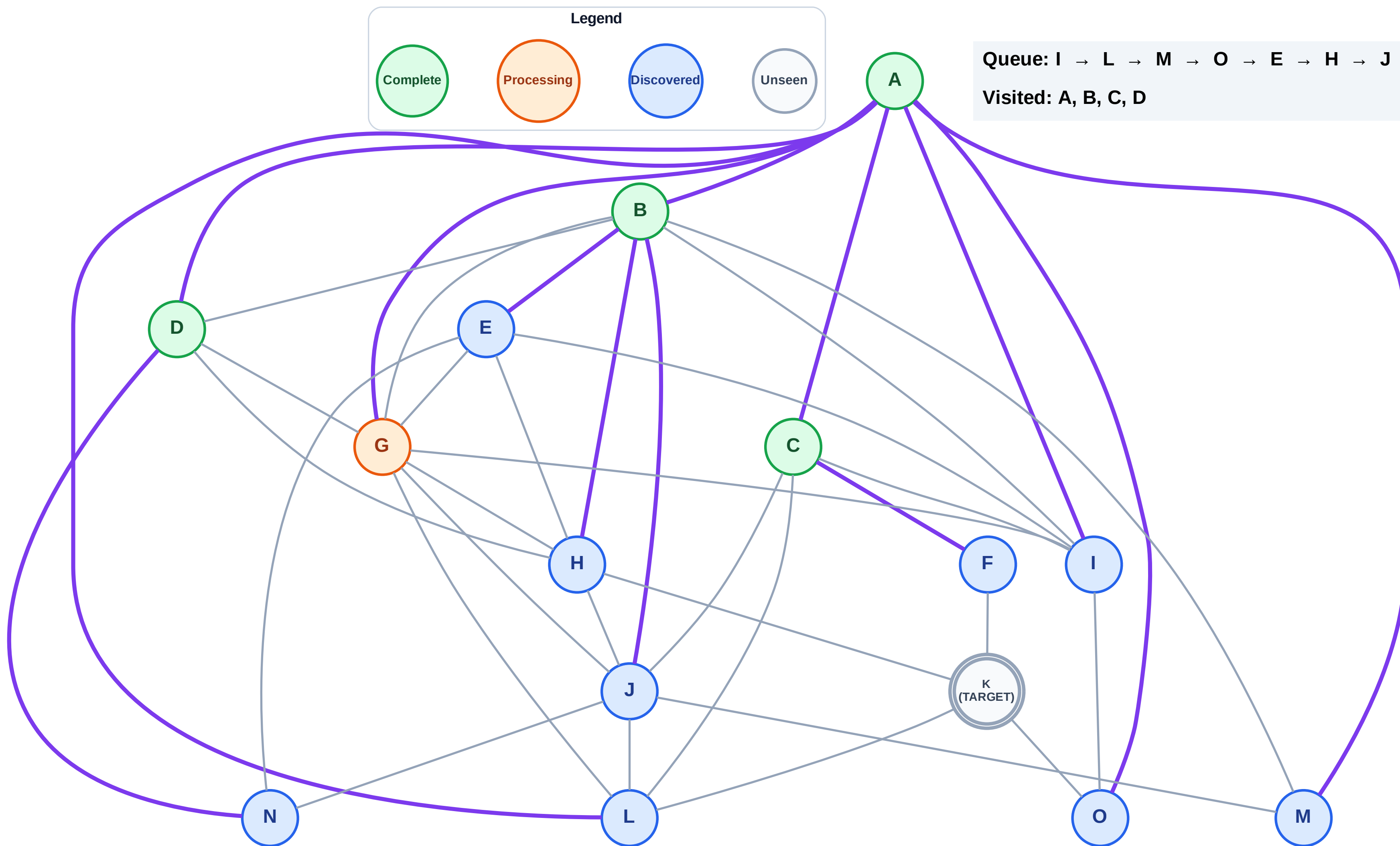
Discovered



Unseen

Queue: I → L → M → O → E → H → J → F → N

Visited: A, B, C, D



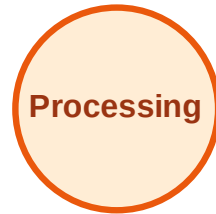
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



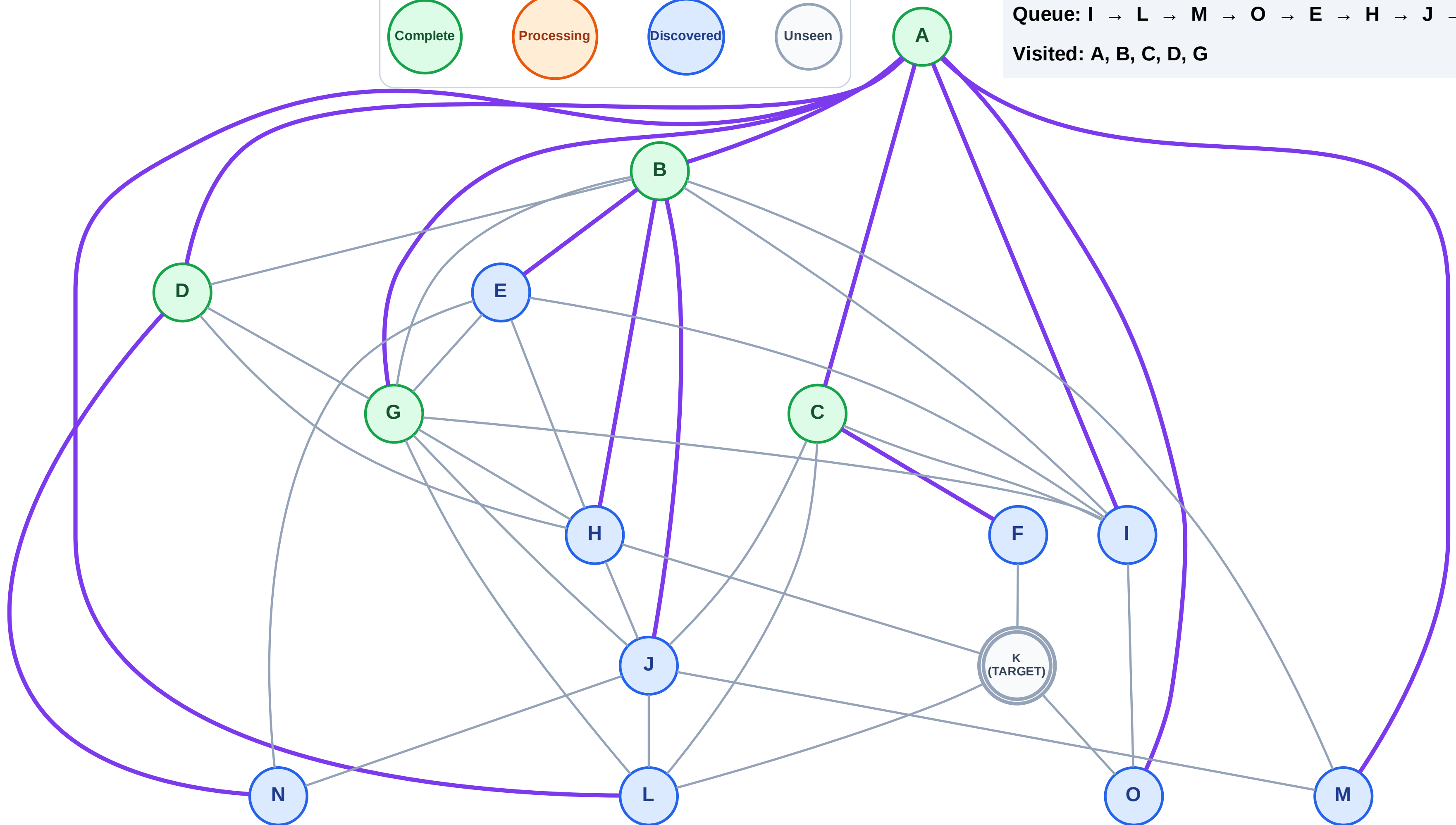
Discovered



Unseen

Queue: I → L → M → O → E → H → J → F → N

Visited: A, B, C, D, G



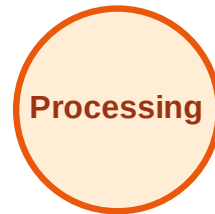
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



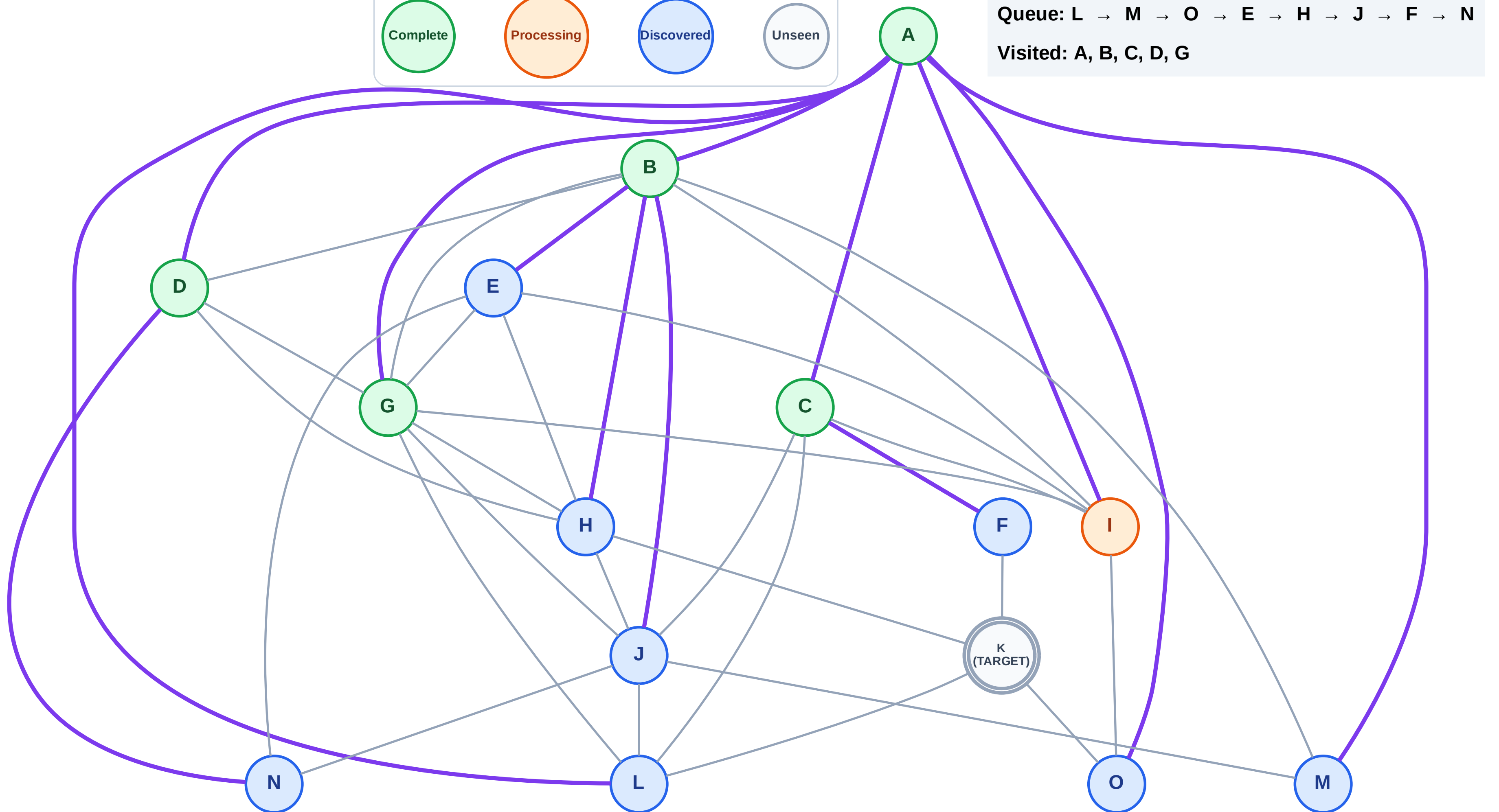
Discovered



Unseen

Queue: L → M → O → E → H → J → F → N

Visited: A, B, C, D, G



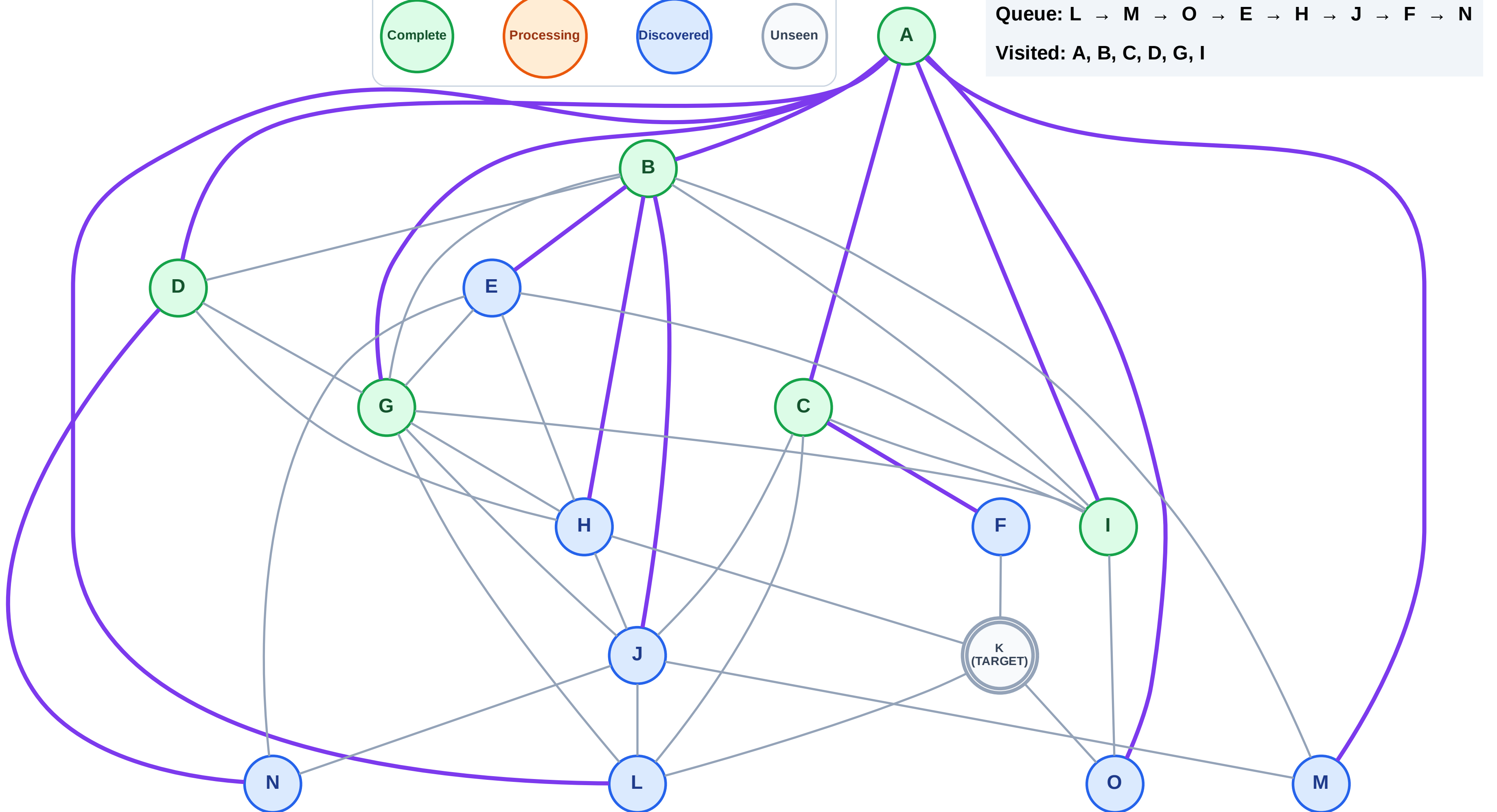
Step 13: No new neighbors from I

Legend



Complete Processing Discovered Unseen

Visited: A, B, C, D, G, I



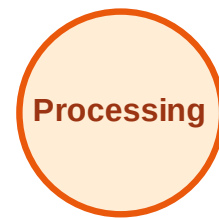
BFS Traversal

Step 14: Process node L

Legend



Complete



Processing



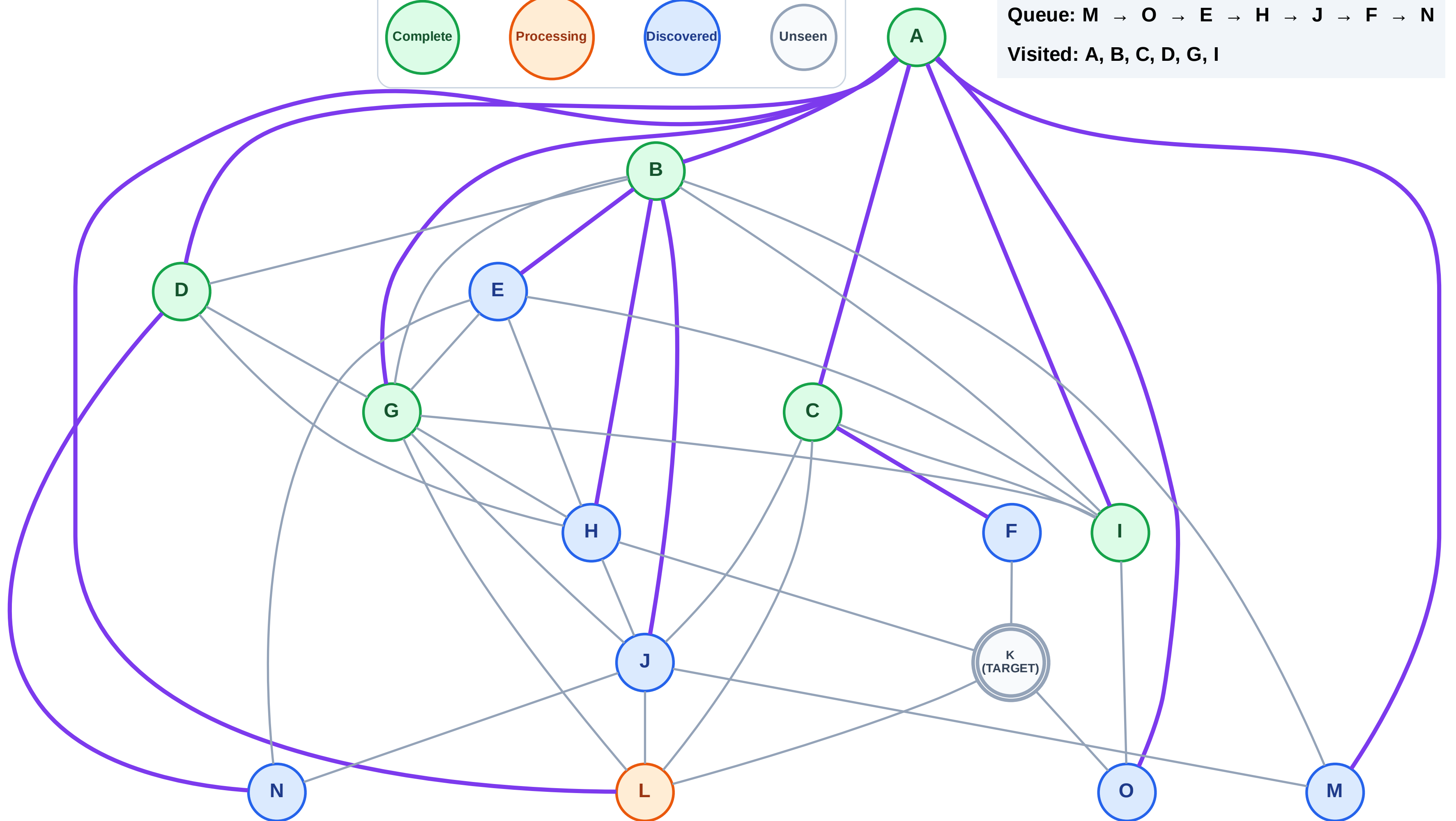
Discovered



Unseen

Queue: M → O → E → H → J → F → N

Visited: A, B, C, D, G, I



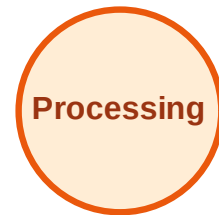
BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → L → K

Legend



Complete



Processing

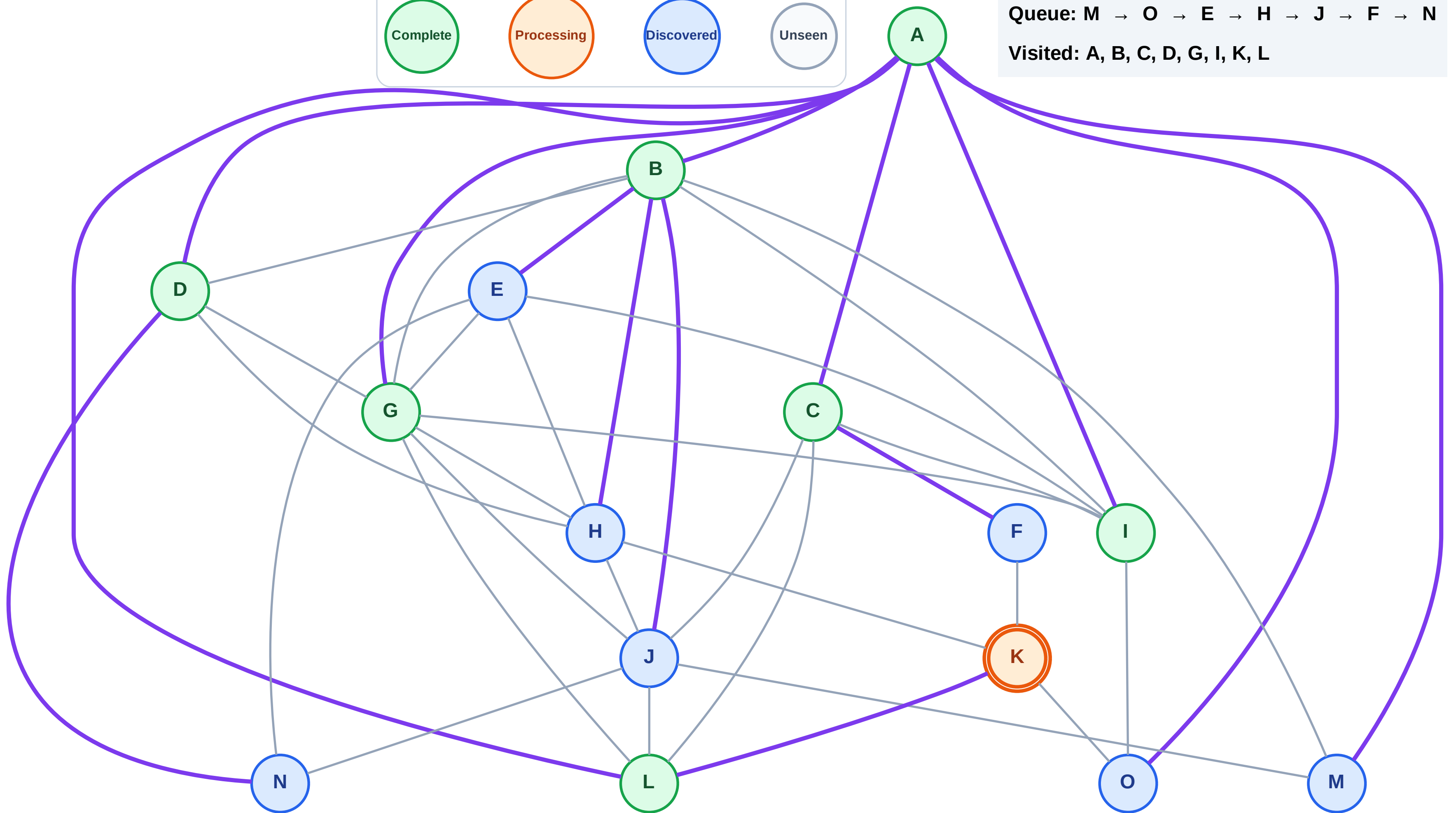


Discovered



Unseen

Queue: M → O → E → H → J → F → N
Visited: A, B, C, D, G, I, K, L



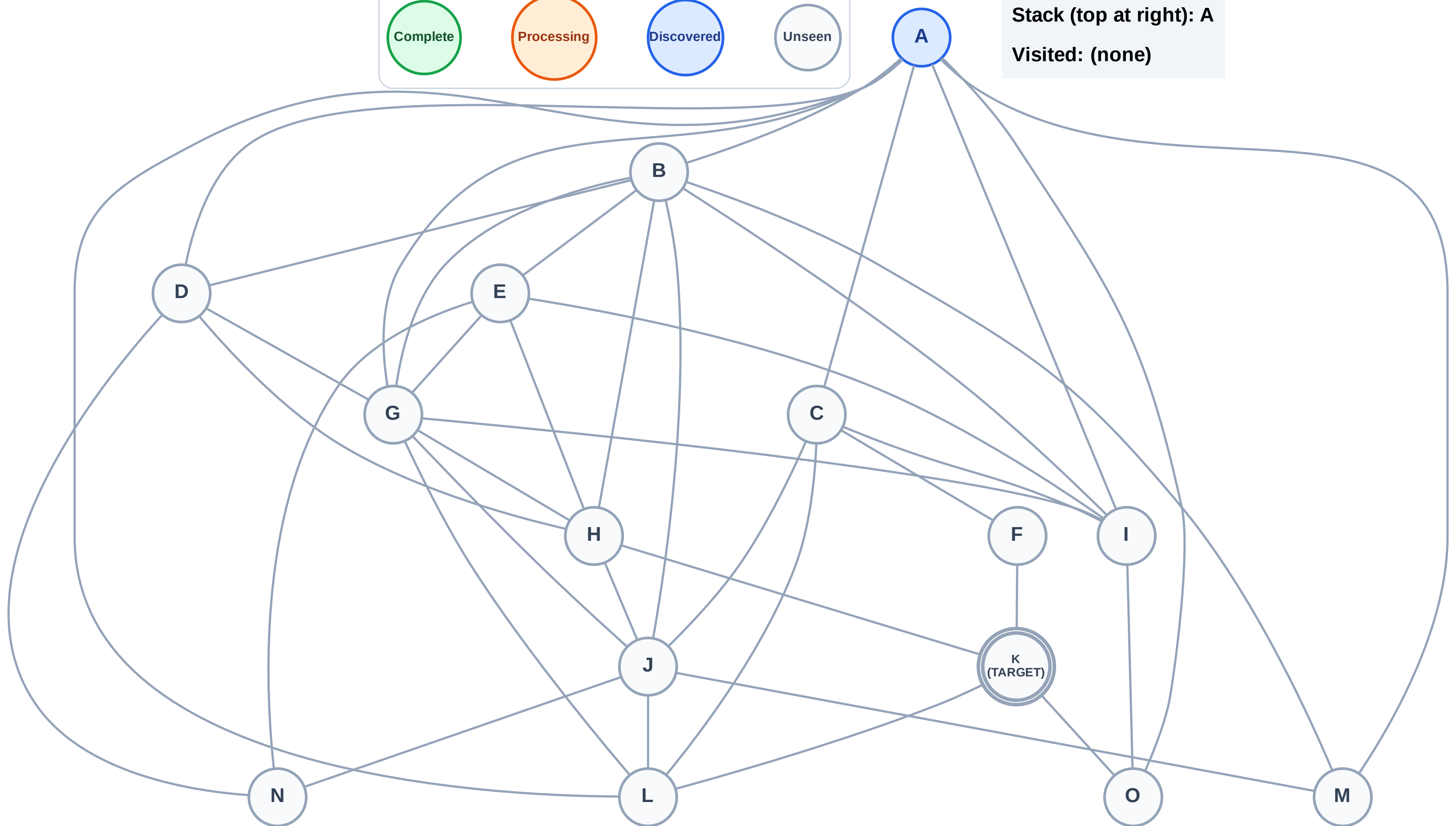
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

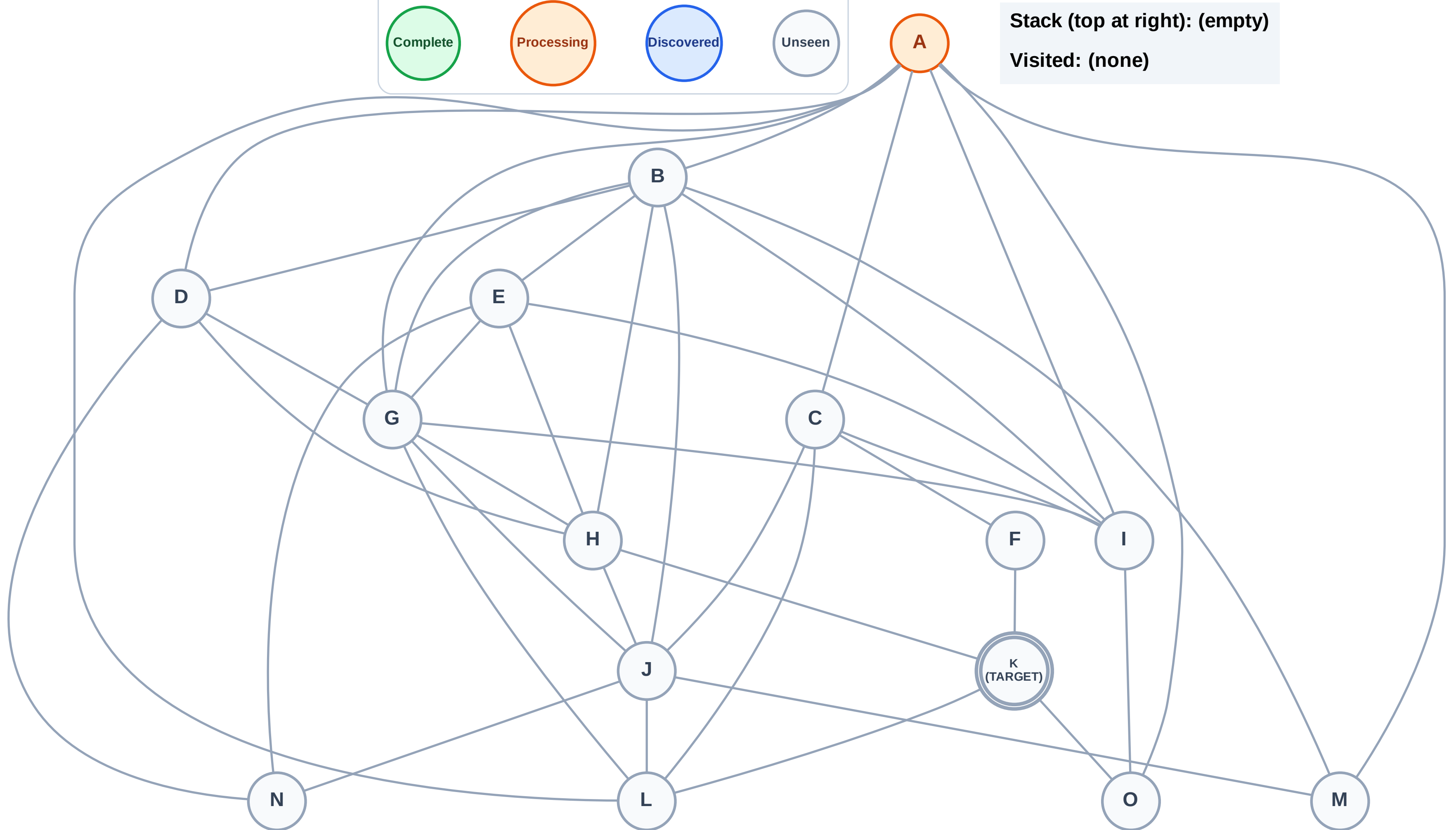
Step 2: Process node A

Legend



Stack (top at right): (empty)

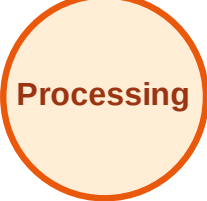
Visited: (none)



DFS Traversal

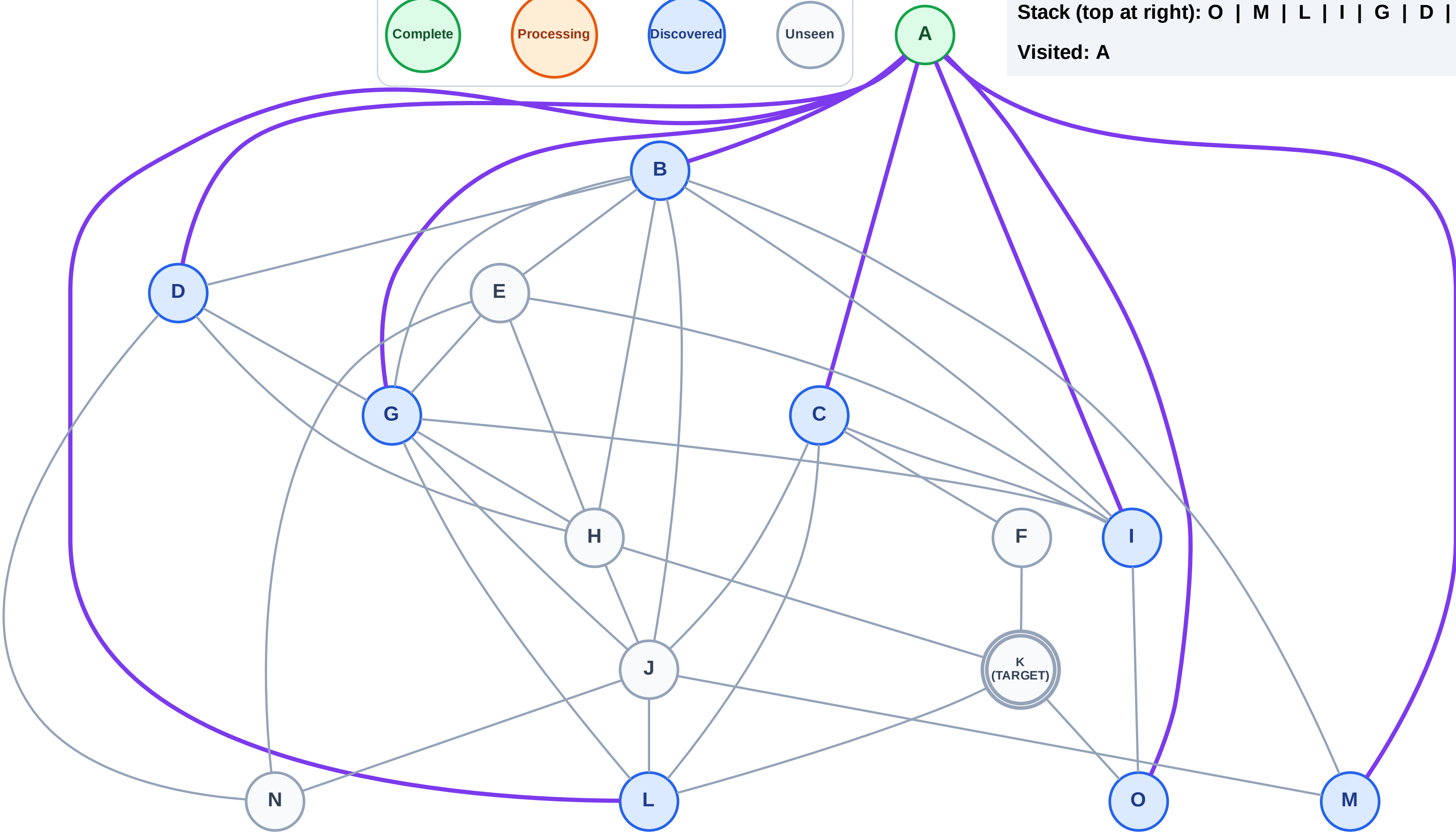
Step 3: Discover O, M, L, I, G, D, C, B from A

Legend



Stack (top at right): O | M | L | I | G | D | C | B

Visited: A



DFS Traversal

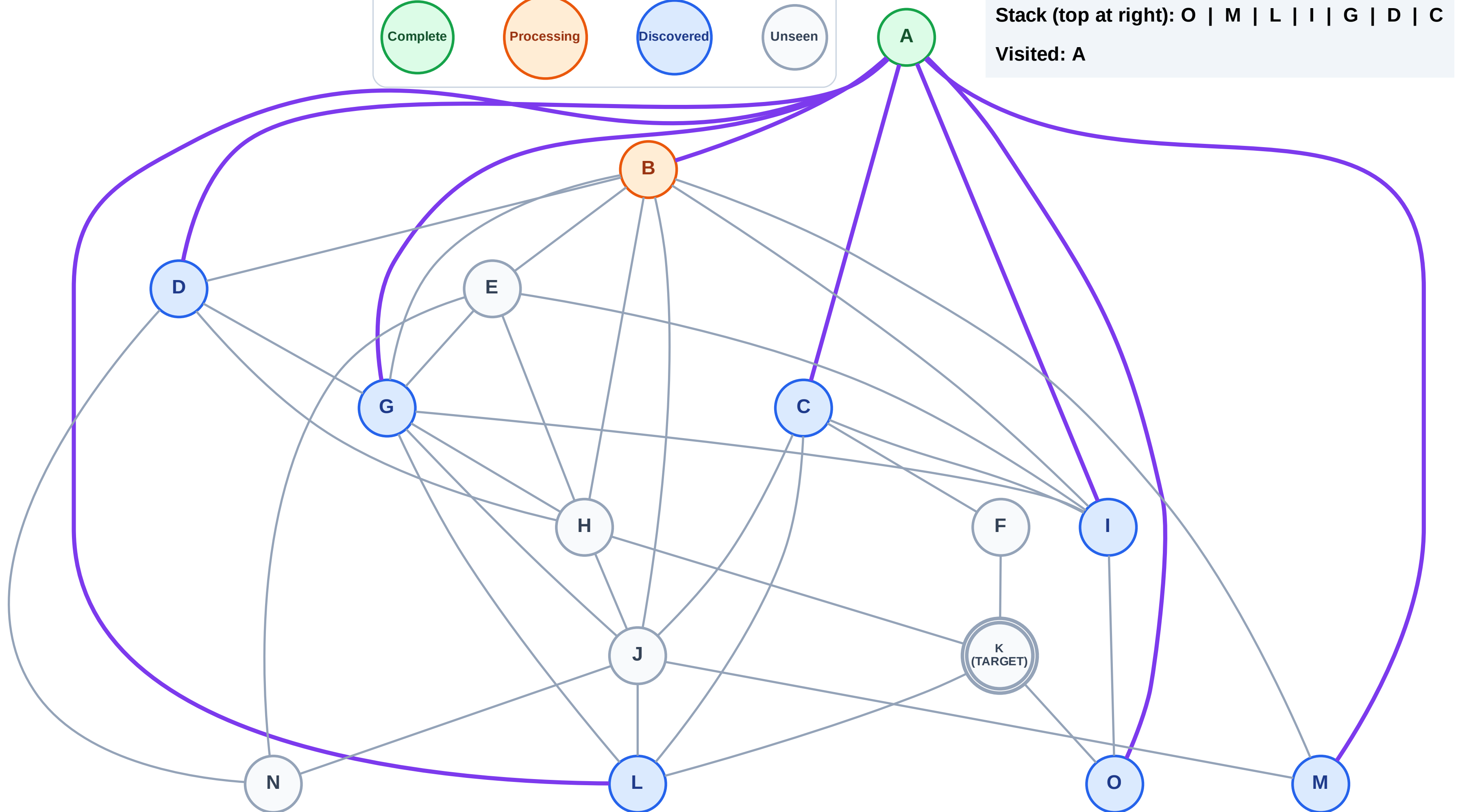
Step 4: Process node B

Legend



Stack (top at right): O | M | L | I | G | D | C

Visited: A



DFS Traversal

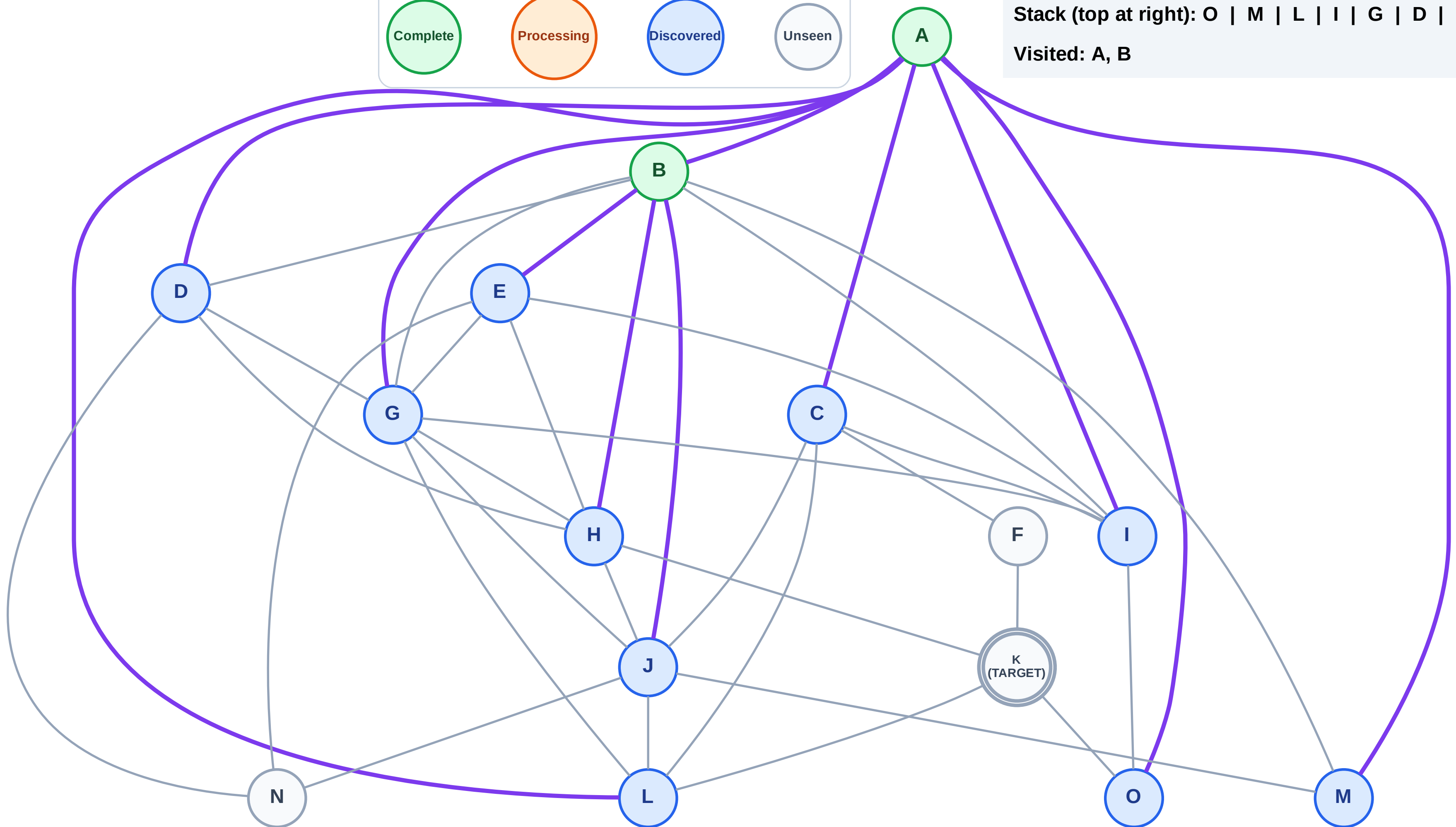
Step 5: Discover J, H, E from B

Legend



Stack (top at right): O | M | L | I | G | D | C | J | H | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

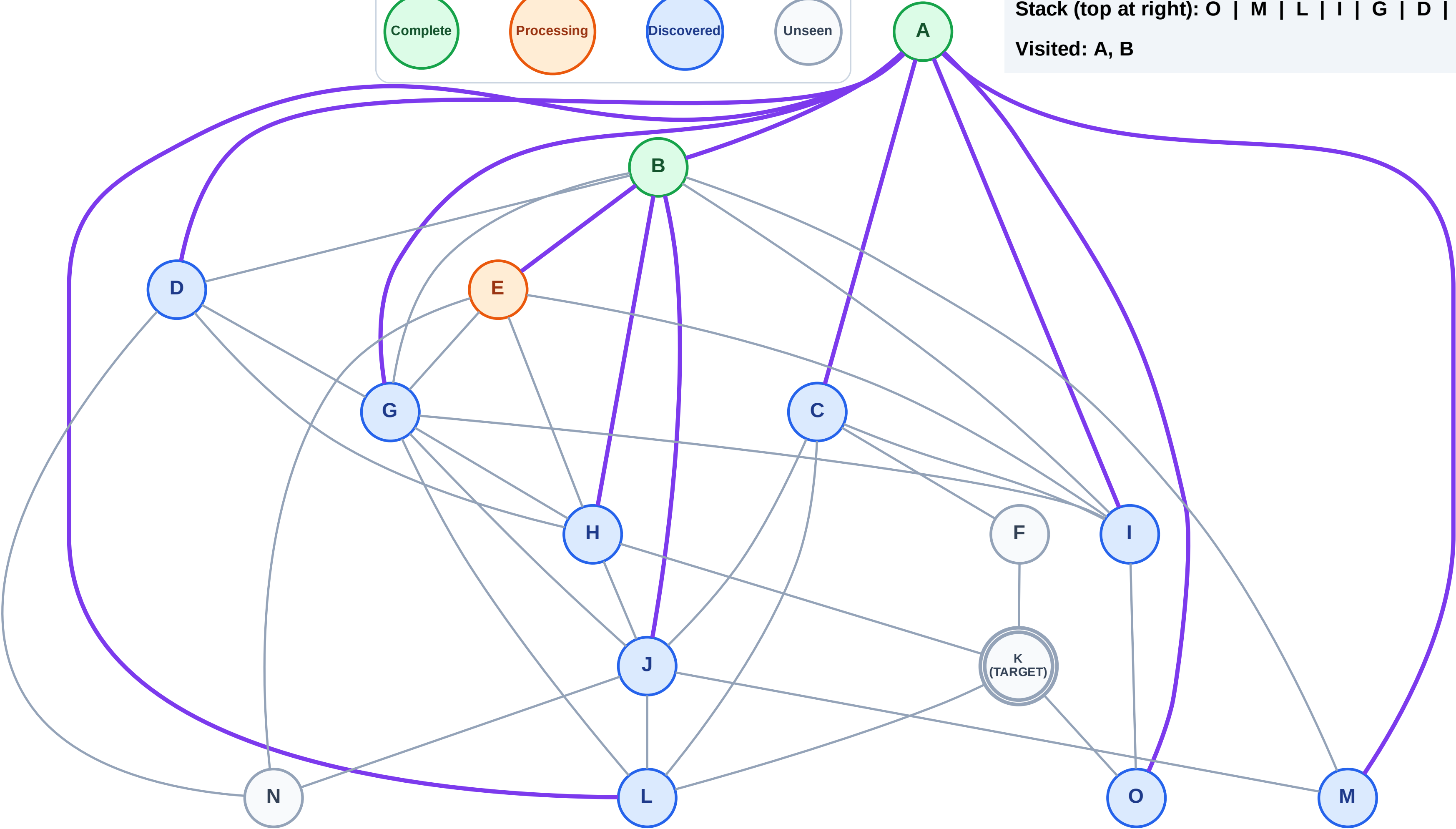
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B



DFS Traversal

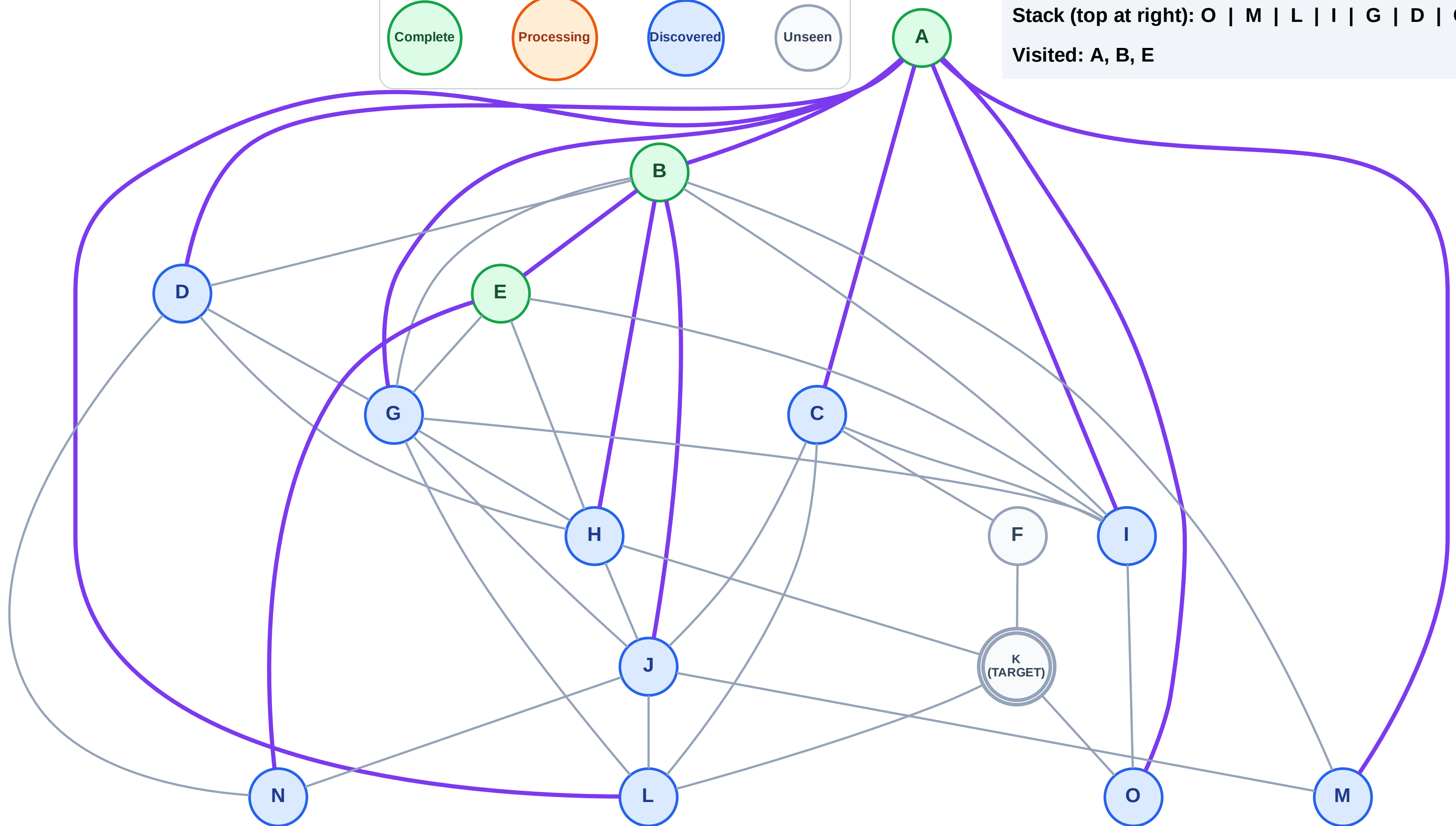
Step 7: Discover N from E

Legend



Stack (top at right): O | M | L | I | G | D | C | J | H | N

Visited: A, B, E



DFS Traversal

Step 8: Process node N

Legend

Complete

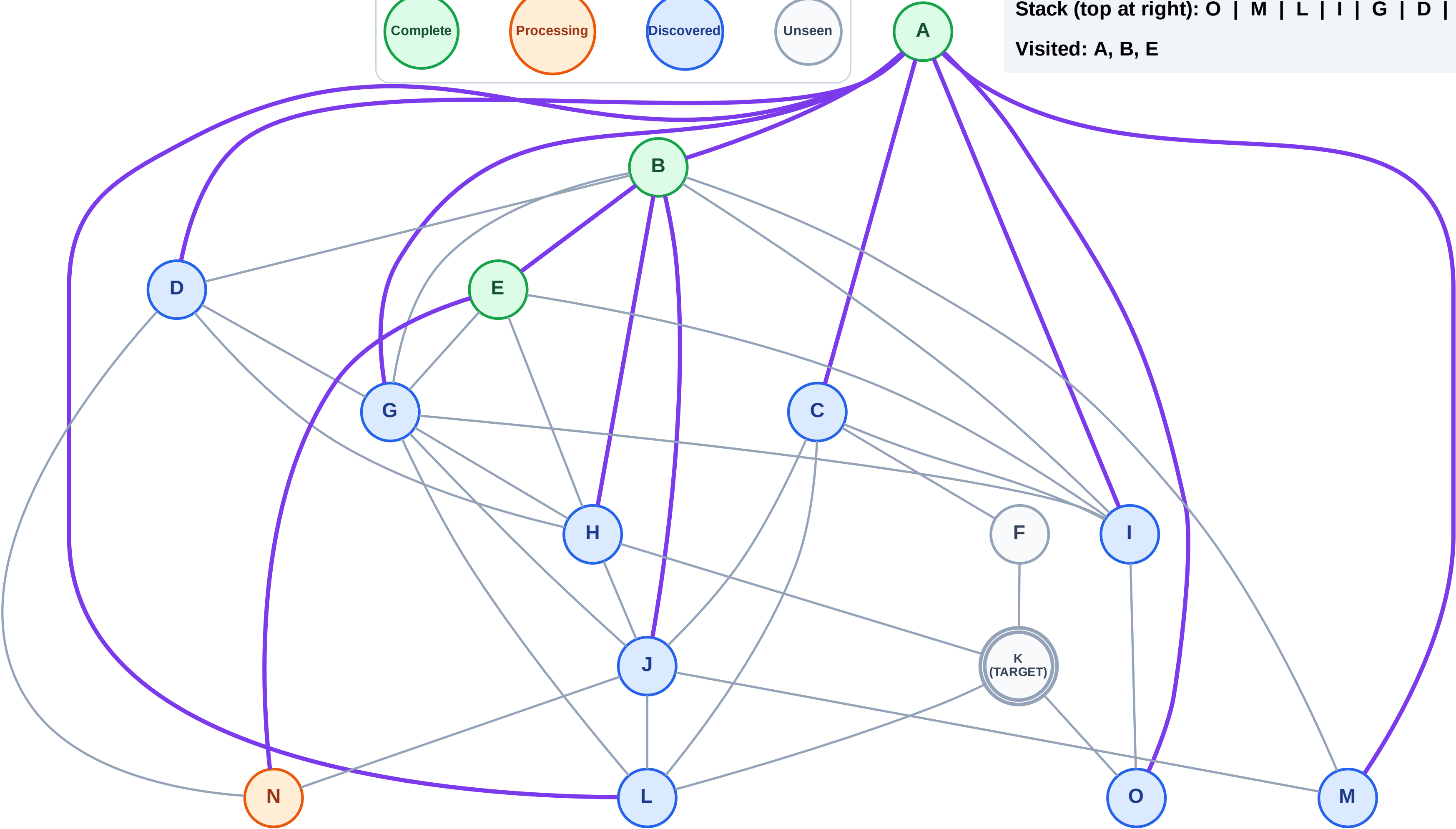
Processing

Discovered

Unseen

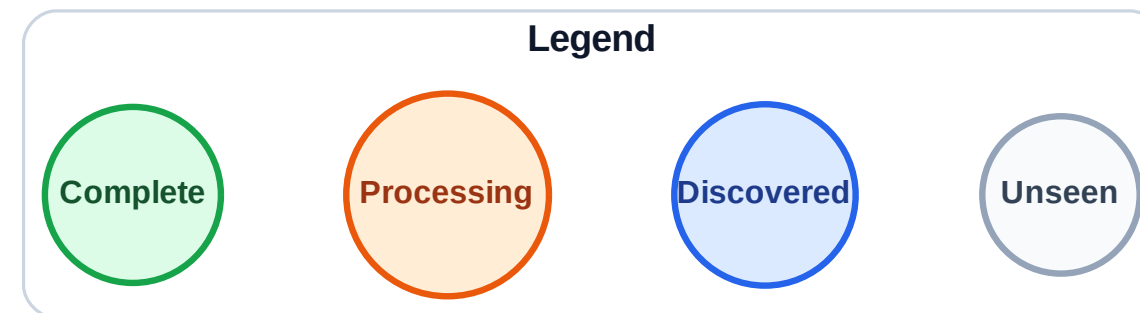
Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B, E



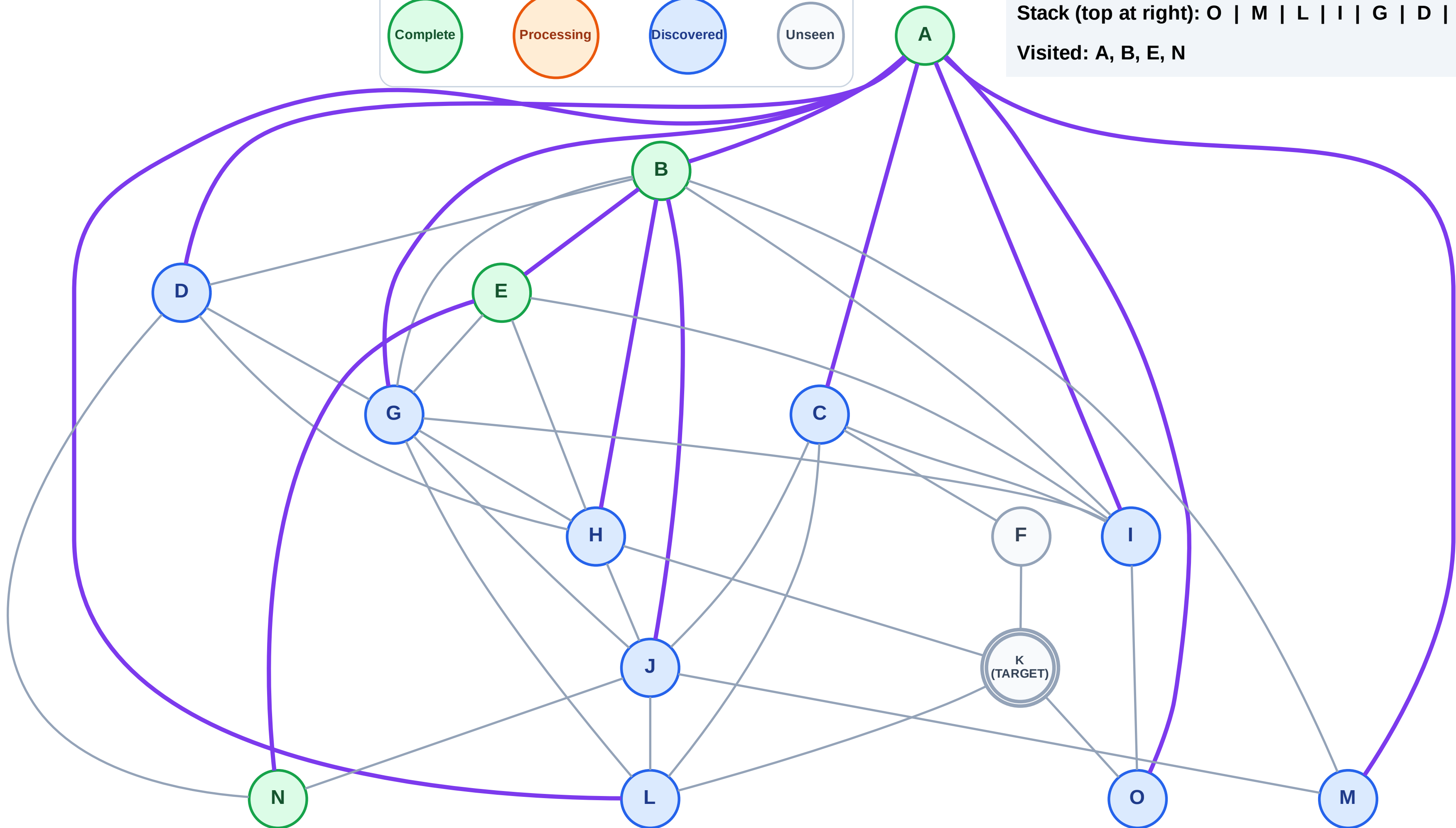
Step 9: No new neighbors from N

Step 9: No new neighbors from N



Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B, E, N



DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



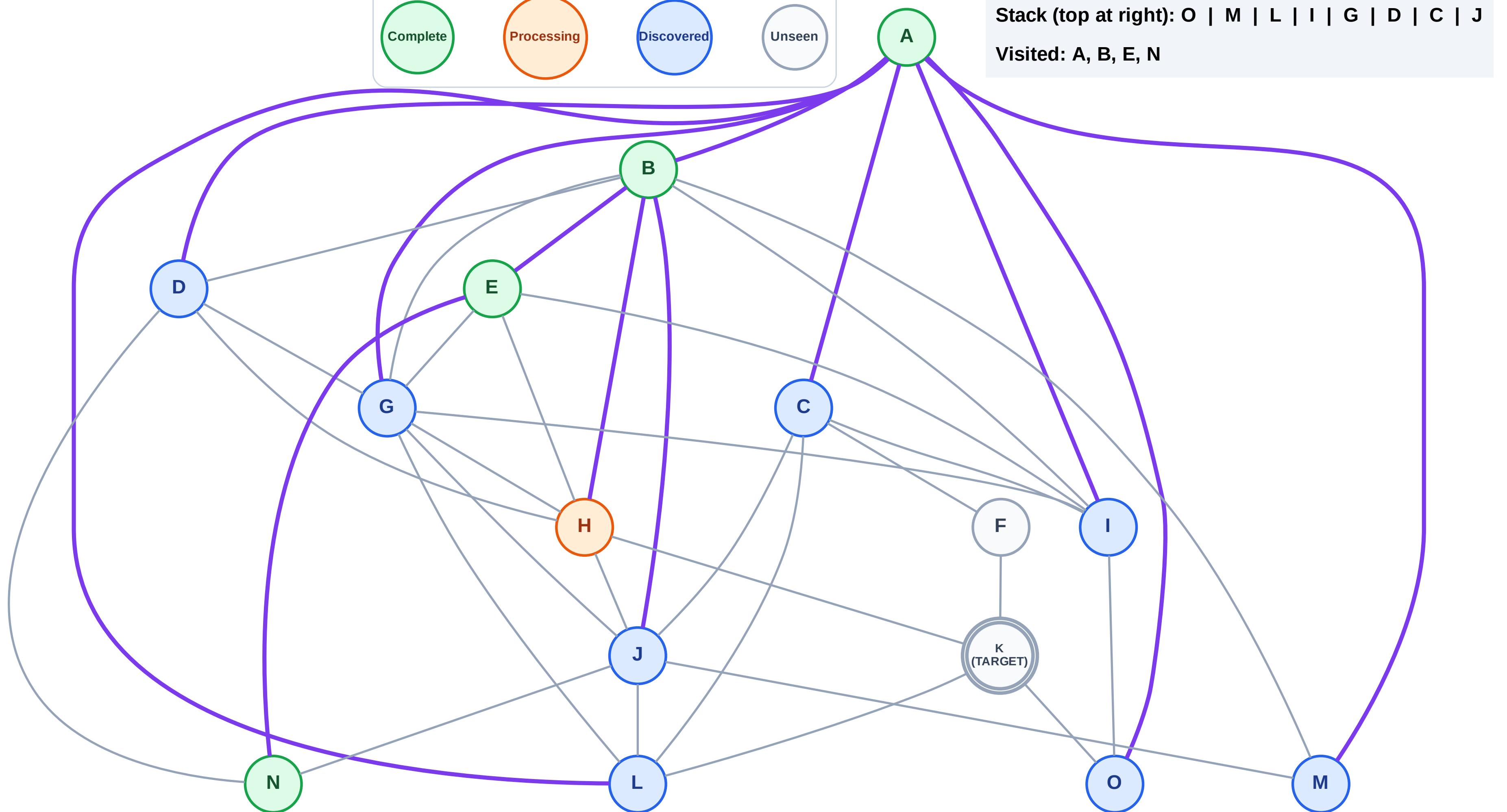
Discovered



Unseen

Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, N



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → H → K

Legend



Processing



Discovered



Unseen



Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, H, K, N

